

MAUS User Guide

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Note: This is a placeholder (“dummy”) guide scaffold. Replace the bracketed [TODO] sections with project-specific content before publishing.

Methyl Assignments Using Satisfiability — a clean-room reimplementaion of the SAT-based methyl NMR assignment method of Nerli et al., *Nat. Commun.* 12:691 (2021).

1. Overview

MAUS assigns methyl NMR resonances by **subgraph isomorphism** rather than by scoring. Given a protein structure and real **peak lists** (HMQC + NOESY), it returns — for each peak — the *set* of methyls consistent with every hard constraint. It never emits a single guess unless the data force one, and it provably never excludes a valid assignment.

Concept	In this tool
Structure graph G	one node per methyl carbon; geminal / short / long edges
Data graph H nodes	HMQC peaks: (¹ H, ¹³ C) shift + residue type
Data graph H edges	NOESY cross peaks, endpoints matched to HMQC by frequency
Solver	python-sat (Glucose3), assumption-based enumeration
Output	per-peak option set (1 / 2–3 / >3 / unassigned)

2. Installation

```
python3 -m pip install python-sat
git clone https://github.com/deepnmr/maus.git
cd maus
```

Requirements:

- Python ≥ 3.8
 - python-sat (provides the Glucose3 SAT solver)
-

3. Quick start

1. build peak lists from a BMRB shift file + the PDB

```
python make_peaklists.py examples/mbp/1ANF.pdb
                        examples/mbp/bmr7114_3.str
```

2. run MAUS on the HMQC + NOESY peak lists (--truth scores against the key)

```
python maus.py examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv
                        examples/mbp/noesy.tsv \
--truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05 --out
                        mbp_options.tsv
```

Expected summary:

```
methyls(G nodes)=192  HMQC peaks=192  NOESY cross peaks=1650
NOE match: firm=502  ambiguous(dropped)=1148  unmatched=0
unique(1 option)      = 51/192
ambiguous(2-3 options)= 81/192
ambiguous(>3 options) = 60/192
unassigned             = 0/192
truth in option set   = 192/192 = 100.0%
```

4. Input formats

4.1 PDB structure

Standard ATOM records. Only these residue types are parsed by default: ALA ILE LEU MET THR VAL. Chain A (or blank) is used.

4.2 HMQC peak list (TSV) — data-graph nodes (input)

label	<TAB> H_ppm	<TAB> C_ppm	<TAB> res_type
P1	0.828	24.510	L
P7	0.340	25.390	L45D2

- label — anonymous peak id (P1, P2, ...); leaks nothing about the answer
- H_ppm, C_ppm — methyl (^1H , ^{13}C) chemical shifts
- res_type — one-letter code (A I L M T V), known from labeling, **or** a *tentative assignment* like L45D2 (Leu45 C δ 2; the leading C is optional, so L45D2 = L45CD2). A tentative cell pins that peak's candidate methyls to the single named one; the constraint then propagates through the NOE network to reduce ambiguity elsewhere. A tentative label absent from the structure is ignored (falls back to the residue type). Runs print tentative anchors used.

Ground truth lives in a **separate** file, never in the input:

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type <TAB> True
P1      0.828      24.510      L      L7CD1
```

Pass it with `--truth` to score; omit it to run blind.

4.3 NOESY peak list (TSV) — data-graph edges

```
label <TAB> C1 <TAB> C2 <TAB> H2
X1      24.56      25.39      0.340
```

3D (H)CCH methyl-methyl cross peak. The **observed** methyl is (H2, C2); the NOE **partner** contributes carbon C1 only (a 3D peak has no partner proton). During the run the observed methyl is matched to an HMQC peak by (H2,C2) and the partner by carbon C1 alone (within `--tol-h/--tol-c`). Single distance class (structure edge $\leq$ long_cut). Because ^{13}C alone is degenerate, most cross peaks match several candidate partners and drop as ambiguous — carbon tolerance is the main resolution lever.

4.3b HMBC-HMQC peak list (TSV, optional) — geminal links

```
label <TAB> C1 <TAB> C2 <TAB> H2
B1      24.56      25.39      0.340
```

Same layout as the NOESY list. One row per Leu/Val residue: observed methyl (H2, C2), geminal partner carbon C1. Pass with `--hmbc`; MAUS forces the pair onto a geminal structure edge. The carbon-only partner is degenerate (on MBP $\sim 1/50$ links resolve), and even a firm link couples the pair without fixing which residue — so per-peak option *counts* are essentially unchanged. The never-exclude guarantee is preserved.

4.4 Generating the peak lists

`make_peaklists.py` builds all three files from a PDB and a BMRB NMR-STAR shift file (`bmrXXXX_3.str`): `hmqc.tsv` (input), `hmqc_true.tsv` (truth key), and `noesy.tsv`. HMQC (^1H , ^{13}C) coordinates are the real BMRB methyl shifts; NOESY cross peaks are emitted for structurally close methyl pairs.

5. Command-line options

Option	Default	Meaning
<code>--short-cut</code>	6.0	structure short-range edge cutoff (Å)
<code>--long-cut</code>	10.0	structure long-range edge cutoff (Å)
<code>--tol-h</code>	0.02	^1H NOESY/HMBC $\rightarrow$ HMQC match tolerance (ppm)
<code>--tol-c</code>	0.20	^{13}C NOESY/HMBC $\rightarrow$ HMQC match tolerance (ppm)
<code>--hmbc</code>	—	optional HMBC-HMQC geminal-link peak list
<code>--truth</code>	—	truth-key TSV for scoring
<code>--labeling</code>	A;I;L;M;T;V	residue types present

Option	Default	Meaning
--out	–	write per-peak options TSV

make_peaklists.py options: --noe-short (6.0), --noe-long (8.0), --keep-k (12 nearest NOE partners per methyl).

6. Interpreting output

The --out TSV columns:

Column	Meaning
label	input peak identifier (anonymous)
res_type	one-letter residue type
n_options	number of valid methyls
options	comma-separated methyl labels
truth	ground-truth label (blank if no --truth)
truth_in_set	1 if truth ∈ options, else 0 (blank if no --truth)

- **1 option** → uniquely assigned.
- **2–3 options** → residual ambiguity, usually geminal/local symmetry.
- **>3 options** → weak local NOE network.
- **0 options** → over-constrained inputs (check cutoffs).

7. On the bundled benchmark

MAUS sees only the two peak lists. HMQC (^1H , ^{13}C) coordinates are the real BMRB 7114 shifts, and each NOESY endpoint is resolved back to an HMQC peak **by frequency**, not by identity — so the assignment is not circular and shift degeneracy has real consequences.

- truth in option set = 100% is the MAUS **guarantee**: a valid assignment is provably never excluded (not a measurement — a property of the exact enumeration).
- The residual ambiguity is **real** and dominated by the 3D format: the NOE partner is matched on ^{13}C only, which is degenerate, so most cross peaks match several candidate partners and drop. Carbon tolerance is the main lever (unique 29 → 51 → 70 / 192 at --tol-c 0.1 / 0.05 / 0.02). A 4D experiment resolving the partner proton would recover far more.

The NOESY cross peaks themselves are still *generated from the structure* (close methyl pairs), so this remains a controlled benchmark rather than a picked experimental NOESY. Swap in a real NOESY peak list to go fully experimental — the maus.py interface does not change.

8. Troubleshooting

Symptom	Likely cause	Fix
ModuleNotFoundError: pysat	solver not installed	<code>pip install python-sat</code>
many unassigned peaks	structure cutoffs too tight	<code>raise --long-cut</code>
most peaks ambiguous	too many NOEs dropped as ambiguous	<code>tighten --tol-h/--tol-c</code>
high unmatched NOE count	tolerance too tight vs shift precision	<code>loosen --tol-h/--tol-c</code>

9. References

- Nerli, De Paula, McShan & Sgourakis. *Backbone-independent NMR resonance assignments of methyl probes in large proteins*. **Nat. Commun.** 12:691 (2021). [doi:10.1038/s41467-021-20984-0](https://doi.org/10.1038/s41467-021-20984-0)
- See `COMPARISON.md` for the head-to-head against MAGIC.

[TODO] Add author, license, and contact sections before release.